

AI-SATHI

Voice-first WhatsApp bot for West Bengal SHG women: bookkeeping, pricing guidance, product catalog creation, and market intelligence, entirely in spoken Bengali.

Production domain: <https://aisathi.app>

WhatsApp webhook: <https://aisathi.app/webhook/whatsapp>

DigitalOcean deployment guide: <docs/digitalocean-deploy.md>

Messaging runs exclusively on the **official Meta WhatsApp Cloud API** - no Twilio, no Baileys, no other third-party messaging provider.

AI vendors, and why each one:

- **Sarvam AI** — sole external AI vendor. [sarvam-30b](#) for standard-tier structured text (ledger extraction, corrections, market phrasing, pricing explanations), [sarvam-105b](#) for higher-stakes phrasing (ad captions, negotiation), [sarvam-vision](#) for catalog photo identification, and [saaras:v3](#) for speech-to-text. **There is no OpenAI dependency anywhere in this codebase** — see <docs/architecture.md> §8.
- **Flux Pro** (optional) — poster-generation upgrade. Leave [FLUX_API_KEY](#) blank and posters are still generated locally via Pillow, free, always.
- **Self-hosted fallback (local Ollama, strongly recommended)** — the *only* fallback tier for every text/vision agent now that OpenAI has been removed. [faster-whisper](#) remains the dedicated free STT fallback regardless of this setting. See <docs/COST.md> for the full cascade table.

What's here

Feature	What it does
Voice-Ledger	Bengali voice note → Banglish/code-mixed normalization (only when needed) → structured income/expense extraction → confirm/correct loop → database write → bank-submittable PDF. Confirmation can be a tap-to-confirm WhatsApp Flow (optional, WA_LEDGER_CONFIRM_FLOW_ID) instead of typed হ্যাঁ/না, removing typo/mishearing risk on the way to a permanent write — see docs/architecture.md §10.3.
Friend-style Pricing Chat	A SELLER-facing conversational back-and-forth ("দর ঠিক করি একসাথে") to agree an asking price before a catalog poster is composed — warm, references seasonal/festival timing, never lets the model state a number, and hard-stops at the seller's own cost-derived floor rather than silently capping below it. See docs/architecture.md §10.4.

Feature	What it does
Catalog Creator	Product photo → background removal → vision product ID (Sarvam Vision, local Ollama vision fallback) → dual Bengali captions (Sarvam, dignity-guideline-constrained) → price suggestion (or the agreed price from the pricing chat above) → optional privacy-respecting market-demand note → composited into a single shareable ad poster (Flux Pro if configured, always falls back to a free local Pillow composite)
Pricing Recommendation	Deterministic price-floor/recommendation math from the seller's own cost, margin, and minimum price (seller_profiles table), blended with market data where available — Sarvam is used only to phrase the explanation, never to generate the number itself
Negotiation	Deterministic accept/reject against the same price floor, code-computed counter-offers (never LLM-generated), a post-generation safety scan that discards any LLM phrasing quoting below the floor, and (as of Pass 4) a deterministically-chosen negotiation <i>tactic</i> (anchor / reciprocity / value-justification / graceful walk-away — shared/knowledge/negotiation_playbook.py) that the LLM only phrases, never picks — see docs/architecture.md §9.1 and §10.4
Cross-agent verification	services/orchestrator/nodes/cross_verify.py — an independent second model call checks tone/dignity, combined with a deterministic check that every ₹ figure in a composed message matches a code-computed value, before the highest-stakes pricing-chat messages are sent. See docs/architecture.md §10.4.
Market Predictor	k-anonymized (min. 5 distinct sellers) aggregation of ledger sales data by block, fused with optional mandi price data and the shared festival/district-mela/seasonal knowledge base, into rising/saturated trend advice
General conversation	Off-topic messages get a real, warm, cheap (Sarvam-routed) reply that gently steers back on-topic

Government Scheme RAG (hallucination-guarded, code-complete but not wired into V3 routing), agri-diagnostics, meeting minutes, training, and subsidy matchmaking are **not** in this build. See [docs/archive/planning/scope.md](#) for the scope-freeze rationale.

Quick start

```
make setup
make dev
```

For a DigitalOcean Droplet:

```
cp .env.example .env
docker compose -f docker-compose.prod.yml up -d --build
```

Postgres schema is applied automatically on first boot
([migrations/0001_init.sql](#), plus additive migrations for the Pricing agent's table and SHG bank-linkage status — [0004_seller_profile.sql](#), [0005_shg_bank_linkage.sql](#)) — no separate migration step needed for a fresh deploy.

Run the test suite any time (no API keys or network required):

```
make test
```

What you must provide

Two things are required to run this at all:

1. **A WhatsApp Cloud API app** (Meta Developer account → WhatsApp product → Phone Number ID + Access Token + App Secret + a verify token you choose yourself). See [SETUP.md](#) for the exact click-through.
2. **A Sarvam AI key** ([sarvam.ai](#)) — this is now the sole paid AI vendor.

Strongly recommended: enable [USE_LOCAL_MODELS=true](#) with a self-hosted Ollama box. With no OpenAI fallback anymore, this is the only thing that keeps every agent alive during a Sarvam outage — see [docs/COST.md](#).

Optional, and safe to leave unset: [WA_LEDGER_CONFIRM_FLOW_ID](#) (tap-to-confirm ledger Flow — falls back to plain text if unset), S3 bucket, Langfuse, mandi price API, Flux Pro, Bengali font for poster generation.

Architecture, in one paragraph

[services/gateway](#) is the FastAPI webhook receiver — it verifies Meta's HMAC signature, deduplicates retried webhooks, rate-limits per number, and hands off to Celery so the 20-second WhatsApp ack window is never at risk.
[services/orchestrator](#) is a LangGraph state machine (Postgres-checkpointed, so every conversation turn is resumable) with one node per agent.
[services/orchestrator/model_router.py](#) is the single place any LLM/vision/translation call goes through — a two-tier Sarvam → local-Ollama cascade for every agent, with retries and hard timeouts, raising a typed [ModelUnavailableError](#) that every node catches and turns into a friendly Bengali message instead of a crash. [shared/knowledge/](#) is the single source of cultural/seasonal/market-timing context (festivals, district melas, life-cycle occasions, negotiation tactics, tone/dignity rules) every agent reads from instead of each node inventing its own.

Full design rationale: see [docs/architecture.md](#) (§8 for the OpenAI removal, §10 for the shared knowledge base / dignity rules / Flow-verified ledger / friend pricing chat / negotiation tactics / cross-verification) and [docs/security.md](#) for what's hardened and why. Step-by-step first-run instructions: see [SETUP.md](#). Turn-by-turn detail on each incremental pass that built §10's features lives in [CHANGELOG_v4_knowledge_and_dignity.md](#) through [CHANGELOG_v9_district_melas.md](#) at the repo root — those files are the "what changed and what's still honestly unverified" record; this README and [docs/architecture.md](#) are the current-state summary.

Repository layout

```

services/
  gateway/          FastAPI WhatsApp webhook + request boundary
                    whatsapp_flows/: scheme_eligibility_flow.json,
                    ledger_confirm_flow.json (tap-to-confirm ledger
entries)
  orchestrator/     LangGraph state machine, feature nodes, model router
                    (Sarvam -> local Ollama cascade, no OpenAI)
    nodes/
      pricing_node.py      Pricing Recommendation agent – deterministic core
      negotiation_node.py  Negotiation agent – code-enforced price floor +
                           deterministic tactic selection
      price_chat_node.py   Friend-style seller pricing chat, pre-poster
      ledger_confirm_flow_node.py Tap-to-confirm Flow consumer
      cross_verify.py      Independent second-pass dignity/numeric check
    translation_service/   Sarvam client (chat, vision, translate, self-hosted
fallback)
    voice_gateway/        Saaras V3 -> self-hosted faster-whisper STT cascade
    pdf_service/          Bank-submittable monthly report generation
(continuity,
                           SHG grading, declaration + sign-off line)
    vision_service/       Catalog image processing, dual captions, ad-poster
composite
                           (flux_poster_client.py: optional Flux Pro tier;
                           poster_composer.py: free Pillow tier, always
available)
    market_service/      k-anonymized market trend aggregation
shared/
  config/ db/ observability/ storage/ whatsapp/
  knowledge/             Shared cultural/market context: festivals, district
melas,
                           life-cycle occasions, negotiation tactics, dignity
rules –
                           the single source every agent reads from
(docs/architecture.md §10)
  i18n/                  Bengali calendar + centralized Bengali digit/number-
word helpers
  assets/fonts/           Bengali TTF for poster text overlay (you provide the
file)

```

migrations/
first boot
tests/unit/
and

Init SQL + additive migrations, applied automatically on

Fast, offline tests for the security-critical, pricing,

knowledge-base logic

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